

Philadelphia Restaurant Database

Integrated SQL Database for Restaurant Benchmarking and Location Analysis

Executive Summary

This project developed an integrated Philadelphia restaurant intelligence database that combines customer perception, premium dining recognition, neighborhood demographics, and location accessibility into a single structured MySQL system. The database was designed to help investors, analysts, and restaurant operators benchmark restaurant positioning, compare premium and non-premium dining signals, and evaluate potential locations for high-end restaurant concepts.

The project integrates four data sources: Yelp restaurant data, Michelin Guide restaurant data, U.S. Census zip-code level demographic data, and Walk Score accessibility metrics. Yelp provides broad market coverage and customer-facing indicators such as ratings, review volume, and price range. Michelin provides an external premium dining signal. Census data adds neighborhood context, including income, population, poverty count, education level, and median age. Walk Score contributes accessibility measures through walk, transit, and bike scores.

The final database includes more than 7,000 Yelp restaurant records, 33 Michelin restaurant records, 53 census records, and restaurant-level accessibility scores. The schema was implemented in MySQL using four core tables: `yelp_restaurants`, `micelin_restaurants`, `census_data`, and `micelin_yelp_match`. Because Yelp and Michelin do not share a common key, a bridge table was created to link matched records while preserving unmatched Michelin restaurants for separate analysis.

The analysis shows that matched Michelin restaurants have higher average Yelp ratings, substantially higher review counts, and higher price ranges than non-Michelin restaurants. Michelin restaurants are also located in areas with stronger accessibility scores, higher median income, higher education levels, and lower poverty counts. These findings demonstrate how a multi-source database can support more informed decisions about restaurant benchmarking, premium market positioning, and location strategy.

1. Business Context

Restaurant performance is influenced by multiple factors, but the relevant information is often fragmented across separate sources. Yelp captures customer sentiment and engagement, Michelin captures premium dining recognition, Census data reflects neighborhood purchasing power and demographics, and Walk Score measures accessibility. A single-source dataset cannot fully explain restaurant market position or support location decisions for premium dining concepts.

This project addresses that gap by building a decision-ready database that connects restaurant-level performance signals with neighborhood and accessibility indicators. The database supports three practical questions:

- How do Michelin and non-Michelin restaurants compare on Yelp performance indicators?
- Are premium restaurants located in areas with stronger accessibility?
- Do premium restaurants cluster in neighborhoods with different demographic profiles?

2. Data Sources and Collection

The project collected and prepared data from four complementary sources. Each source contributes a different perspective on restaurant performance and market positioning.

Source	Role in the Database	Key Fields
Yelp Open Dataset	Primary restaurant dataset and broad market coverage	<code>business_id</code> , <code>name</code> , <code>address</code> , <code>postal_code</code> , <code>latitude</code> , <code>longitude</code> , <code>stars</code> , <code>review_count</code> , <code>price_range</code> , <code>is_open</code>

Source	Role in the Database	Key Fields
Michelin Guide	Premium dining benchmark collected through web scraping	name, postal_code, latitude, longitude, price_level, cuisine, distinction
U.S. Census API	Neighborhood demographic context by zip code	population, median_income, poverty_count, bachelor_degree, median_age, postal_code
Walk Score API	Location accessibility metrics at the restaurant level	walkscore, transit_score, bike_score

2.1 Data Collection and Cleaning

- Converted Yelp JSON files into structured tabular files and retained restaurant-level fields relevant to benchmarking and database integration.
- Collected Census demographic variables by iterating through Philadelphia restaurant postal codes and using conditional API handling to skip unavailable zip codes.
- Scraped Michelin restaurant listings and parsed combined text fields such as price and cuisine using regular expressions and string cleaning.
- Collected Walk Score metrics using restaurant addresses and coordinates, then embedded accessibility scores directly into the Yelp and Michelin restaurant tables.
- Standardized postal codes, restaurant names, and identifier fields to improve consistency across sources.

3. Database Design and Integration

The database was designed as a four-table MySQL schema that balances normalization, analytical usability, and data integration constraints. Restaurant-level accessibility scores are embedded in the Yelp and Michelin tables because they vary by restaurant. Census variables are stored separately because they are zip-code level attributes shared by many restaurants.

Table	Purpose	Primary Key / Linking Logic
yelp_restaurants	Stores restaurant operating attributes, Yelp performance indicators, Michelin match flag, and accessibility scores.	restaurant_id; business_id retained as source identifier
michelin_restaurants	Stores premium dining attributes and accessibility scores for Michelin-listed restaurants.	michelin_id; postal_code links to census_data
census_data	Stores zip-code level demographic variables used for neighborhood analysis.	postal_code
michelin_yelp_match	Bridge table connecting Michelin records to matched Yelp restaurants while preserving unmatched Michelin records.	michelin_id; restaurant_id when a Yelp match exists

3.1 Integration Logic

Yelp records use business_id values that are not directly compatible with Michelin records. To support relational joins and bridge-table integration, the project created numeric restaurant_id and michelin_id fields. Michelin and Yelp restaurants were matched using cleaned restaurant names: lowercased names with surrounding whitespace removed.

The project intentionally used exact name matching rather than fuzzy matching. This reduced the risk of incorrectly linking different restaurants with similar names. The trade-off is that some true matches may be missed when names differ slightly across sources. For this use case, match precision was prioritized over match coverage because incorrect matches would weaken the credibility of premium restaurant comparisons.

3.2 Design Trade-offs

- Accessibility scores were embedded in restaurant tables because they are restaurant-specific and frequently used in comparisons.
- Census attributes were kept in a separate table because they are zip-code level values shared by multiple restaurants; separating them avoids redundant updates and improves maintainability.
- A bridge table was used because Yelp and Michelin do not share a common identifier, and some Michelin restaurants do not appear in the Yelp dataset.
- Exact name matching was selected to reduce false positives, even though it may under-match restaurants with small naming differences.

4. Results and Business Value

4.1 Analysis 1: Michelin vs. Non-Michelin Yelp Performance

The first analysis used the `is_michelin` flag in the Yelp table to compare restaurants that were successfully matched to Michelin records against the broader non-Michelin restaurant population.

Restaurant Group	Total Restaurants	Avg. Stars	Avg. Review Count	Avg. Price Range
Non-Michelin	7,128	3.58	99.17	1.62
Michelin (matched)	17	4.32	615.18	2.93

Matched Michelin restaurants show stronger Yelp performance across all three indicators. They have higher average ratings, substantially higher review volume, and higher average price ranges. This suggests that Michelin recognition is associated with stronger customer engagement and premium positioning among the matched restaurants.

4.2 Analysis 2: Accessibility of Michelin vs. Non-Michelin Restaurants

The second analysis compared accessibility scores for all Michelin restaurants against non-Michelin Yelp restaurants. This analysis used restaurant-level Walk Score, Transit Score, and Bike Score fields.

Restaurant Group	Total Restaurants	Avg. Walk Score	Avg. Transit Score	Avg. Bike Score
Non-Michelin	7,128	86.29	77.70	77.53
Michelin	33	96.21	82.85	88.64

Michelin restaurants have higher average accessibility scores than non-Michelin restaurants. This indicates that premium dining locations in Philadelphia are more likely to be concentrated in highly walkable, transit-accessible, and bike-friendly areas.

4.3 Analysis 3: Neighborhood Characteristics

The third analysis joined restaurant records to census_data using `postal_code` to compare the neighborhood profiles of Michelin and non-Michelin restaurants. Records without matching Census zip-code data were excluded from this comparison.

Restaurant Group	Total Restaurants	Avg. Median Income	Avg. Poverty Count	Avg. Bachelor Degree Count	Avg. Median Age
Non-Michelin	7,064	\$79,816.17	6,085.45	6,049.84	34.53
Michelin	33	\$94,755.30	3,677.33	7,689.76	34.69

Michelin restaurants are located in zip codes with higher median income, higher education levels, and lower poverty counts. Median age is similar across the two groups. These patterns can help analysts and operators identify neighborhood characteristics associated with premium restaurant positioning.

4.4 Business Applications

- Benchmark restaurants against premium dining signals using ratings, review volume, price range, and Michelin status.
- Evaluate location attractiveness using accessibility and demographic indicators.
- Identify neighborhoods that may be better suited for high-end restaurant concepts based on income, education, and accessibility profiles.
- Provide a reusable relational database foundation for future dashboards, predictive modeling, or market-entry analysis.

5. Limitations and Future Improvements

- The number of Michelin restaurants in Philadelphia is small, which limits the statistical strength of Michelin vs. non-Michelin comparisons.
- Not all Michelin restaurants could be matched to Yelp records, which reduces coverage for Yelp-based performance comparisons.
- Exact name matching improves precision but may miss true matches when names differ slightly across sources.
- Yelp metrics reflect online engagement and may not fully represent revenue, profitability, or broader customer sentiment.
- Future work could expand the project to comparable cities, add Google Reviews or foot traffic data, and use fuzzy matching with manual review for improved match coverage.

6. Individual Contributions

Team Member	Primary Contribution Area	Specific Contributions
Jasjyot Singh	Michelin and Yelp data collection	Scraped Michelin data, downloaded Yelp data, and converted source data formats.
Riya Elizabeth	MySQL data insertion and reporting support	Inserted prepared datasets into MySQL tables and contributed to the executive summary and limitations section.
Adam Li	Census API and database design documentation	Collected Census data through API requests and documented data sources, schema design, and integration logic.
Wenjing Duan	Walk Score API, data cleaning, SQL tables, and analysis	Collected Walk Score data through API requests, cleaned data, created MySQL tables, wrote SQL queries, and summarized results and business value.

Appendix: Repository Files

The full project repository is intended to include the following supporting files:

- philadelphia_restaurant_database.ipynb - Python workflow for data collection, cleaning, API calls, matching logic, and database loading.
- schema.sql - MySQL database schema and table creation statements.
- analysis_queries.sql - SQL queries used for Michelin vs. non-Michelin comparisons and business analysis.
- restaurant_database_report.pdf - Polished project report for review and portfolio use.